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Predicting Standby (IPB) Passenger Show-Ups for KLM Operations

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Abstract



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Abstract

The abstract should summarize the contents of the thesis. It should be clear, descriptive, self-explanatory and not longer than a third of a page. Please avoid using mathematical formulas as much as possible. Keywords might be given.

Keywords: Forecasting, Optimization.

1 Introduction

What is inflation forecasting and why is it important?

Inflation is a persistent increase in the general price level of goods and services within an economy over a period of time. It is considered one of the most important key economic indicators. The future inflation rate can heavily affect households, economic agents, authorities, and policy makers in making optimal consumption decisions and investment strategies over time. High inflation rates weaken economic growth rates, decrease real wages and savings, and raise production costs, while low rates represent a low demand in the economy, as indicated by less and less spending and consumption and the duration of investment projects. Central banks usually rely on inflation predictions to implement effective and powerful economic policies and to anchor their expected inflation rates to make the economy more stable. Therefore, producing accurate inflation forecasts plays a crucial role in economic decision-making. This research aims to compare the traditional time series models and machine learning (ML) models in inflation forecasting.

Why is inflation forecasting very difficult?

Inflation forecasting is a difficult task. This mainly stems from the presence of high correlation, nonlinearity, and high dimensionality between the macroeconomic and financial variables and even the potential structural breaks in the data, challenging the traditional forecasting techniques and the formation of a forecasting equation ([Aras and Lisboa \(2022\)](#), [Doojav and Narmandakh \(2025\)](#)). However, the combination of the recent advancements of ML models and the availability of big datasets appears to be a potential alternative solution to these issues.

Research gap in previous studies

Most of the previous inflation forecasting literature widely agreed that the multivariate time-series models outperformed the univariate ones in inflation forecasting. However, the usage and benefits of ML models combined with large datasets have been scarcely discussed in these studies in the field of macroeconomic forecasting. In particular, One of the seminal studies is [Atkeson et al. \(2001\)](#) evaluates the usefulness of the Phillips curve-based models for inflation forecasting based on real economic activity, but eventually, the authors argued that the search for other augmented Phillips-curve models should be abandoned due to their lower forecast accuracies compared to naive models and the rising complexity of inflation in the United States over the past 15 years. On the other hand, the study of [Stock and Watson \(1999\)](#) verifies that the Phillips curve with different

specifications yields more reliable and stable inflation forecasts compared to univariate forecasting models, but their work is limited by using only linear models and assuming a linear relationship between inflation and the candidate variables, and no forecasting horizon longer than 1 year is used to evaluate the models. As a result, the existing research on inflation forecasting is inadequate to model the complex relationship between covariates and inflation, leading to inaccurate predictions that can deteriorate the effectiveness of the central bank’s policy.

In recent years, due to the advancements in computing power and the availability of larger datasets, ML models have gained more and more attention, and economists and researchers have considered them to be potential alternative approaches to the classical time-series models to enhance the inflation forecasting accuracy. Several studies have been conducted to verify this finding. One of them is [Nakamura \(2005\)](#) showed that on average, the neural networks outperform univariate autoregressive models for short horizons when forecasting U.S. inflation. Another finding of [Garcia et al. \(2017\)](#) revealed that high-dimensional econometrics models have excellent performance in real-time inflation forecasting within a data-rich environment in Brazil, confirming the usefulness of big data in macroeconomic forecasting. However, [Medeiros et al. \(2021\)](#) raised a question about whether similar findings of this study can be replicated for the case of the U.S. economy. [Medeiros et al. \(2021\)](#) with a comprehensive analysis found that ML models incorporating a large number of covariates systematically outperform the univariate benchmark models in terms of inflation forecasting accuracy, and the results of this paper also confirm the superiority of random forest models to predict U.S CPI inflation rates. There are numerous studies that show a preference for ML techniques combined in a data-rich environment. For instance, the empirical work of [Doojav and Narmandakh \(2025\)](#) indicates that ML approaches such as XGBoost, Random Forest and Ridge Regression with large datasets produce more accurate inflation forecasts than AR and factor models. This paper also emphasizes the strength of ML models in handling nonlinearities, multicollinearity, and dimensionality. Another result of [Huang et al. \(2025\)](#) showed that ML methods based on a large number of macroeconomic and financial predictors are superior in China’s inflation forecasting compared to conventional time series, especially during periods of high inflation volatility. This paper also highlights that penalized linear regression models, including ridge and elastic net, are worth considering given their outstanding inflation-forecast performance. A narrow replication of [Naghi et al. \(2024\)](#) confirms the results of [Medeiros et al. \(2021\)](#) On the dominance of Random Forest for US inflation forecasting, this finding is not universal when considering extended sets of models and US data, where other competing tree-based models with better forecast performance surpass RF.

Research Aims

Given the lack of research regarding inflation forecasting, this study aims to compare and evaluate the inflation forecasting performance of benchmark models, including two univariate (AR, RW and factor models) and two multivariate (VAR and BVAR) time series models and machine learning models (Random Forest, Lasso and Ridge regression and adapt Lasso, Elastic Net, Bagging) across different forecast horizons within a data-rich environment using the same monthly data and the updated version for the U.S economy and examine the importance of ML approaches in forecast-

ing U.S. inflation and to verify the results and findings of the [Medeiros et al. \(2021\)](#). To do so, this study fills these gaps by first simulating inflation and predictors. Second, different feature selection techniques are employed and evaluated in reducing the number of covariates and identifying the most influential independent features, and this relates to a previous paper by [Aras and Lisboa \(2022\)](#), who proposed the usage of feature selection in a high-dimensional environment. Then, multiple models are tested at different forecast horizons using the simulated sample and the selected candidate variables and later applied to the real FRED-MD data, which is the monthly macroeconomic database compiled by [McCracken and Ng \(2016\)](#).

Explain why my research is worth doing and the value/ benefit it will provide to academia and industries

This study will contribute to the inflation forecasting literature, which can be summarized as follows. Firstly, it may generalize on new and unseen datasets, preventing the potential overfitting issue during the model training and testing process, since there are only a few studies that compare and test models on simulated data first. Secondly, this work can better identify the most informative predictors that are potentially defined as the key drivers of U.S. inflation dynamics and determine the contribution of each selected feature to the prediction problem ([Aras and Lisboa \(2022\)](#)). The final contribution is to provide a comprehensive analysis and assessment of the ML models in inflation forecasting over different horizons. These findings can further contribute to enhancing the inflation forecasting accuracy and, therefore, help the central banks to achieve optimal monetary policies.

The potential limitations of my project and approach

However, there are several limitations. Although this study uses a simulated dataset to replicate the data-rich environment for the model training and testing process, this cannot completely guarantee that my findings can be generalized to other countries due to the different natures of each economy, causing different volatility patterns in inflation rates; hence, achieving desired accuracy in inflation forecasting is no longer possible.

Structural outline

The rest of this study is constructed as follows. Section II gives an overview of the literature. Section III describes the methodology involving traditional econometrics and ML models. Section IV introduces the data, hyper-parameter tuning. Section V provides the main results in evaluating and comparing the forecasts of competing models. Section VI is the conclusion. The appendix provides additional results and figures.

2 Methodology

Forecasting Framework

This section describes the methods and models used in this study to forecast future U.S. CPI inflation. Our main goal is to forecast the inflation rate π_{t+h} at period $t + h$ using the information

set available at period t . A direct multi-step-ahead forecasting framework is considered for future inflation because this approach is more robust to model misspecification, in contrast to the iterated or recursive forecasting method, which is constructed iteratively using one-step-ahead model forecasting model's estimators (Marcellino et al. (2006)). Other reasons for choosing the direct forecast approach are to avoid the necessity of estimating a model for the time evolution of predictors x_t and to circumvent the issues of specification errors in the recursive settings (Garcia et al. (2017) & Huang et al. (2025)). We consider the following inflation forecasting model

$$\pi_{t+h} = F_h(x_t) + u_{t+h}, \quad h = 1, \dots, H, \quad t = 1, \dots, T.$$

Where π_{t+h} is the inflation at month $t + h$, where T denotes the full sample size and h is the forecasting horizon, 1-,3-,6-, 12-, and 24-month ahead forecasts are considered in this paper. $\mathbf{x}_t = (x_{1t}, \dots, x_{nt})'$ is an $(n \times 1)$ vector of predictor variables, containing lags of inflation π_t , a large set of macroeconomic and financial explanatory variables up to time t , common factors computed from a large set of potential covariates and lags of factors. $F_h()$ is a function mapping the predictors $\mathbf{x}_t$ to the future inflation π_{t+h} , u_{t+h} is a zero-mean random error. The target function $F_h(x_t)$ can be a single model or an ensemble of different models, and there is a different functional mapping for each forecasting horizon (Medeiros et al. (2021) & Doojav and Narmandakh (2025)).

The direct multi-step ahead forecasting approach is based on the rolling-window schemes of fixed-length. This rolling procedure not only reduces the impacts of potential structural breaks and outliers, but it also helps capture the model specification uncertainty, model instability, and estimation uncertainty, and the uncertainty of future events during the model estimation process using the data up to date t (Medeiros et al. (2021) & Huang et al. (2025)). This is indicated through the following direct forecasting equation form

$$\hat{\pi}_{t+h|t} = \hat{F}_{h, t-R_{h+1:t}}(\mathbf{x}_t).$$

Where $\hat{F}_{h, t-R_{h+1:t}}$ is the estimated target function using the data spans from the time $t - R_h + 1$ until time t and R_h is the size of the window, which varies based on the forecasting horizon and the number of lagged variables considered in the model. However, the in-sample size depends on the forecasting horizon h and the number of lags p where the number of in-sample observations is computed as $R_h = M - h - p - 1$, and M stands for the total number of months available in the dataset up to time t (Medeiros et al. (2021)).

Forecasting inflation is constructed using different models ($F_h(\mathbf{x}_t)$), including benchmark models (AR, Random Walk, UCSV, AORW) and supervised Machine Learning models, containing the penalized regression models (Ridge, LASSO, Elastic Net, Addaptive LASSO), tree-based models (Random Forest/RF, and extreme gradient boosting (XGBoost)), and additional advanced ML models such as Neural Network, GAS model, and Bootstrap Aggregating (Bagging) method across different forecasts horizons as mentioned above. These models will be described in the section below.

3 Models

3.1 Benchmark Models

In this subsection, the set of benchmark models for inflation forecasting is considered using the work of [Medeiros et al. \(2021\)](#). The AORW model is also added to this set of models. We denote π_t the month-on-month inflation at time t .

The first benchmark model is the Random Walk model (RW). For the forecast horizons $h = 1, \dots, 24$, the RW forecast of period $t + h$ inflation made at time t is $\hat{\pi}_{t+h|t} = \pi_t$, which is the last observed value of inflation.

The second benchmark model is the autoregressive (AR) model of order p or the AR(p) model. Although the lag order p is usually determined by the Bayesian Information Criterion (BIC) for each window, we follow [Medeiros et al. \(2021\)](#) to fix the lag length at four. This fixed order of 4 lags is also considered in the paper of [Huang et al. \(2025\)](#). The parameters are estimated by OLS. The forecast equation is as follows:

$$\hat{\pi}_{t+h|t} = \hat{\beta}_{0,h} + \hat{\beta}_{1,h} \pi_t + \dots + \hat{\beta}_{p,h} \pi_{t-p+1}.$$

Where $\hat{\pi}_{t+h|t}$ represents the h -month ahead forecasted inflation given the information available up to date t . The parameters $\hat{\beta}_{i,h}$ for $i = 0, 1, \dots, p$ are OLS estimates and $\pi_t, \dots, \pi_{t-p+1}$ are the lagged values of inflation. This time-series model assumes a linear relationship between the future inflation rate, the target variable, and its p -lagged values.

The third benchmark is the UCSV model defined by the following equation:

$$\pi_t = \tau_t + e^{h_t/2} \varepsilon_t, \quad \tau_t = \tau_{t-1} + u_t, \quad h_t = h_{t-1} + v_t.$$

Where $\{\varepsilon_t\} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$ is a sequence of independent and normally distributed random variables. u_t and v_t are also normal with mean zero and variances with inverse gamma priors $\tau_1, h_1 \sim \mathcal{N}(0, 0.12)$. This model is estimated by Markov Chain Monte Carlo (MCMC) methods. The direct h -steps-ahead forecast is obtained by $\hat{\pi}_{t+h} = \hat{\tau}_{t|t}$.

The final benchmark is the AO-RW forecasting model, which is essentially a variant of the Random Walk forecast considered by [Atkeson et al. \(2001\)](#). Following the findings of [Stock and Watson \(2007\)](#), we extend the AO forecast to other horizons with any adjustments, particularly by taking the average inflation over the last 12 months. Therefore, the AO forecast is defined as follows:

$$\pi_{t+h|t}^h = \pi_t^{12} = \frac{1}{12} (\pi_t + \pi_{t-1} + \dots + \pi_{t-11}),$$

$$\hat{\pi}_{t+h|t} = \pi_t.$$

This benchmark univariate model is also employed in the studies of forecasting inflation in China of [Huang et al. \(2025\)](#) and in the case study of Brazil [Araujo and Gaglianone \(2023\)](#).

3.1.1 Generalized Autoregressive Score Model (GAS(1,1))

The final benchmark is the GAS(1,1) model, an observation-driven time series model introduced by [Creal et al. \(2013\)](#) and [Harvey \(2013\)](#). The main idea behind this model is to update the parameter over time. This is achieved by the scaled score of the likelihood with a chosen scaling matrix, where the score defines the steepest ascent direction for improving the model's local fit. Applying this to the US CPI inflation forecasting framework, we formulate the model specification as follows,

Let π_t be the dependent variable of interest at time t , the time-varying parameter here is set as the mean f_t of inflation, the variance is assumed to be constant σ^2 , and $\theta = (\omega, A, B, \sigma^2)$ is a vector of static parameters. The reason behind this setup is that this study concentrates on predicting the average inflation level in % instead of inflation volatility; therefore, the conditional mean f_t is chosen to move over time, and the variance is fixed, and the lag orders are fixed at $p = 1$ and $q = 1$ to have a parsimonious model.

We have the distribution of the target variable π_t given the available information $\mathcal{F}_{t-1}$ is Gaussian with a conditional mean and constant variance.

$$\pi_t | \mathcal{F}_{t-1} \sim \mathcal{N}(f_t, \sigma^2), \quad \pi_t = \log(P_t) - \log(P_{t-1}), \quad t = 1, \dots, T$$

The observation density and the corresponding loglikelihood function are therefore given by

$$\begin{aligned} \pi_t &\sim p(\pi_t | f_t, \mathcal{F}_{t-1}, \theta) \\ \ln p(\pi_t | f_t, \mathcal{F}_{t-1}, \theta) &= \ell_t = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(\pi_t - f_t)^2}{2\sigma^2} \end{aligned}$$

The updating equation for the GAS(1,1) model is

$$f_{t+1} = \omega + As_t + Bf_t$$

where ω is a constant, A and B are coefficients. The recursion is initialized at the in-sample mean of each window $f_1 = \bar{\pi}$. Then the time-varying parameter f_t is updated to the next period $t + 1$ with the scaled score, where the score is defined as the derivative of the loglikelihood with respect to the mean f_t

$$\nabla_t = \frac{\partial \ell_t}{\partial f_t} = \frac{\pi_t - f_t}{\sigma^2}$$

The scaling matrix S_t is set equal to the inverse information matrix

$$S_t = I_{t|t-1}^{-1} = \sigma^2, \quad I_{t|t-1} = \mathbf{E}_{t-1}[\nabla_t^2] = \frac{1}{\sigma^2}$$

The scaled score is the driving mechanism of the GAS(1,1) model; in this case is defined as the forecast error.

$$s_t = S_t \cdot \nabla_t = \sigma^2 \cdot \frac{\pi_t - f_t}{\sigma^2} = \pi_t - f_t$$

It is now clear to see that the choice of inverse information matrix of $S_t = \sigma^2$ simplifies the raw score ∇_t to the simple forecast error term $\pi_t - f_t$, avoiding the chance of dividing by a tiny value of variance σ^2 , potentially leading to numerical instability. The maximum likelihood approach is used to estimate the static parameters

$$\hat{\theta} = (\hat{\omega}, \hat{A}, \hat{B}, \hat{\sigma}^2) = \arg \max_{\theta} \sum_{t=1}^T \ell_t$$

Replacing s_t with the obtained forecast error, we have the recursion for the GAS(1,1) model becomes:

$$f_{t+1} = \omega + A(\pi_t - f_t) + Bf_t = \omega + A\pi_t + (B - A)f_t$$

This can be interpreted as an error-correction average inflation where there is a current forecast for the average inflation f_t for each month t . This current forecast is then corrected and updated via the forecast error, and this continues until the end of the sample.

3.2 Penalized Regression Models

Penalized regression is also known as the shrinkage or regularization techniques. These methods introduce a linear regression model with penalties for having too many predictors in the model that can overfit the data. When there are many correlated predictors, or when the number of explanatory variables N is much higher than the number of observations T , overfitting can be considered as a serious problem. In this case, although the in-sample model predictive performance is accurate, its out-of-sample forecasts are inaccurate. To deal with this issue, regularization proposing penalty terms, can eliminate the less contributive variables by reducing the coefficient values of the less-relevant predictors close to or equal to zero.

Given the direct forecasting equation $\pi_{t+h} = F_h(\mathbf{x}_t) + u_{t+h}$, we have $F_h(\cdot)$ is in the format of a linear regression model where

$$F_h(x_t) = \beta'_h \mathbf{x}_t = \beta_{0,h} + \sum_{i=1}^N \beta_{i,h} x_{it}$$

These models minimize the following general penalized regression defined as follows:

$$\hat{\beta}_h = \arg \min_{\beta_h} \left[\sum_{t=1}^{T-h} (\pi_{t+h} - \beta'_h x_t)^2 + \sum_{i=1}^N p(\beta_{h,i}; \lambda, \omega_i) \right]$$

Where T is the total number of observations and N is the number of predictors, $p(\beta_{h,i}; \lambda, \omega_i)$ corresponds to a penalty function that depends on the shrinkage parameter λ and on a weight $\omega_i > 0$. The value of the hyperparameter λ determines the magnitude of the penalty term and the bias-variance tradeoff. A higher value of λ means a greater shrinkage of the regression coefficients β_h , resulting in more bias and less variance in the forecasting model. For the shrinkage-based models, the set of predictors is standardized before estimation.

We consider different choices for the penalty functions $p(\beta_{h,i}; \lambda, \omega_i)$. In this study, we consider four main shrinkage models for forecasting CPI inflation, including the least absolute shrinkage and selection operator (LASSO), adaptive LASSO (adaLASSO), ridge regression, and Elastic Net models.

3.2.1 Ridge Regression (RR)

Ridge Regression is proposed by [Hoerl and Kennard \(1970\)](#). The objective of RR is to deal with highly correlated regressors by reducing the variance of the estimator in exchange for a small bias. RR shrinks the regression coefficients of variables with a tiny contribution to the outcome to nearly zero. This is achieved by penalizing the regression models with an L2-norm penalty term, which is defined below:

$$\sum_{i=1}^N p(\beta_{h,i}, \lambda, \omega_i) := \lambda \sum_{i=1}^N \beta_{h,i}^2$$

RR has the advantage of an analytic solution due to easy computation. However, the coefficients obtained by ridge regression are very small but never exactly equal to zero, regardless of any value of λ . This means that Ridge is not appropriate for feature selection. An interesting fact is that RR is a special case of the Elastic Net if $\alpha = 0$ in the penalty function (1).

3.2.2 The Least Absolute Shrinkage and Selection Operator (LASSO)

The LASSO regression model is an alternative method proposed by [Tibshirani \(1996\)](#). The core idea is to shrink the coefficients of irrelevant variables exactly to zero, thereby overcoming the problem of ridge regression by imposing the penalty function called the L1 norm on the regression coefficients $\beta_{h,i}$, defined as follows:

$$\sum_{i=1}^N p(\beta_{h,i}; \lambda, \omega_i) := \lambda \sum_{i=1}^N |\beta_{h,i}|$$

In this case, LASSO can also perform as a variable selection technique. LASSO can handle the case when there are more variables N than observations T by identifying the correct subset of relevant features. Nonetheless, LASSO also has limitations according to [Zou and Hastie \(2005\)](#); it may be an inappropriate method for choosing features in some cases. One of them is the grouping effect, which happens when the regression coefficients of a group of highly correlated variables tend to be equal. Another case is when $T > N$ when the number of observations is larger than the number of highly correlated predictors, RR tends to outperform LASSO. LASSO is also a special case of the Elastic Net, considering $\alpha = 1$ in the ElNet penalty equation (1).

3.2.3 adaptive LASSO (adaLASSO)

The adaLASSO regression model, proposed by [Zou \(2006\)](#), aims to reduce the estimation bias, achieve variable selection consistency, and improve forecasting accuracy, compared to the standard LASSO, with inconsistency for selecting variables under certain circumstances. AdapLasso uses the

same penalty term as LASSO, but includes an additional weighting parameter ω_i . The core idea behind the model is to use the previously known information to select variables more effectively. AdapLASSO is a two-step estimation procedure that first generates different weights ω_i for each candidate variable $x_{i,t}$, and these resulting weights are then used in the second step of LASSO estimation as additional information. The penalty function is defined as follows:

$$\sum_{i=1}^N p(\beta_{h,i}; \lambda, \omega_i) := \lambda \sum_{i=1}^N \omega_i |\beta_{h,i}|$$

Where $\omega_i = |\tilde{\beta}_{h,i}|^{-1}$ represents the weights and $\tilde{\beta}_{h,i}$ is the respective LASSO coefficient estimated in the first step. Hence, adapLASSO can assign a higher penalty for zero coefficients and a lower penalty for nonzero ones and deal with many more candidate variables N than observations T , and performs well even under non-Gaussian and heteroskedastic errors. When implementing this model, we set $\omega_i = \frac{1}{|\tilde{\beta}_{h,i}| + \frac{1}{\sqrt{T}}}$ to avoid zero denominator as the adapLASSO would set the coefficients of variables exactly to zero and adding the term $\frac{1}{\sqrt{T}}$ would prevent division by zero $\frac{1}{|\tilde{\beta}_{h,i}|}$ (Medeiros and Mendes (2016), Araujo and Gaglianone (2023) and Huang et al. (2025)).

3.2.4 Elastic Net (ElNet)

The Elastic Net model is a generalization including RR and LASSO as special cases, and is proposed by Zou and Hastie (2005). Particularly, it is a regression model penalized by both the L1-norm and L2-norm penalty functions. This resulting combination leads to an effective shrinkage of coefficients towards zero (as in RR) and some coefficients being set to exactly zero (as in LASSO). The ElNet penalty function is defined as follows:

$$\sum_{i=1}^N p(\beta_{h,i}; \lambda, \omega_i) := \alpha \lambda \sum_{i=1}^N \beta_{h,i}^2 + (1 - \alpha) \lambda \sum_{i=1}^N |\beta_{h,i}| \quad (1)$$

Where $\alpha \in [0, 1]$ is the hyperparameter to determine the relative weights of the two penalty terms, L1 and L2 norms. ElNet benefits from the useful properties of Ridge and LASSO simultaneously, namely, shrinkage and sparsity. Ridge regularization allows ElNet to handle many highly correlated predictors, while L1-norm regularization takes advantage of sparsity for the algorithm (Huang et al. (2025) and Aras and Lisboa (2022)).

The above regularized linear regression models (Ridge Regression, LASSO, adapLASSO, and Elastic Net) cannot capture nonlinear relationships between a target variable and the predictors. Therefore, nonlinear ML models, Random Forest (RF), and Extreme Gradient Boosting (XGBoost) are considered in this U.S. inflation forecasting analysis.

3.3 Tree-based ML Models

3.3.1 Random Forest (RF)

The Random Forest (RF), introduced by [Breiman \(2001\)](#), is an ensemble learning model that consists of decision trees used for both regression and classification problems. The main idea behind it is to enhance the predictive accuracy and alleviate the overfitting problem that a regression tree frequently faces, and to reduce its high prediction variance by using the bootstrap aggregation (bagging) of randomly constructed regression trees. In turn, a regression tree is a nonparametric model based on the recursive binary partitioning of the covariate space X . The model is usually displayed in a graph that has the format of a binary decision tree with the root node as the base of the tree, parent nodes or split nodes and terminal nodes, also called leaves, representing different partitions of X ([Araujo and Gaglianone \(2023\)](#) and [Garcia et al. \(2017\)](#)). The procedure of the Random Forest model can be summarized in the following steps. Firstly, multiple decision trees are trained on bootstrapped samples of the data. Each regression tree uses a random subset of features at each node. As a result, the trees are as different as possible since they learn different data patterns and use different predictor subsets. The final output is obtained by taking the average of the individual tree predictions. According to [Medeiros et al. \(2021\)](#) and [Huang et al. \(2025\)](#), the forecasting framework for Random Forest prediction for inflation can be formalized as follows:

First, B distinct bootstrap samples are drawn from the original data. Then, a random subset of the feature vector $\mathbf{X}_t$ is selected for each sample $b = 1, 2, 3, \dots, B$. The decision tree uses one of the chosen variables to split the feature space into two regions at $X_1 = s_1$ and further partitions the covariate space into more non-overlapping regions R_k , where $k = 1, 2, \dots, K$ and each region corresponds to a terminal node of the decision tree. We then repeat the above-mentioned process until the terminal nodes or leaves are too tiny to be split further. The block bootstrap is utilized in this study due to the time series nature in the dataset.

Given the dependent variable π_{t+h} , feature set X_t , and the number of terminal nodes K , each terminal node R_k with $1 \leq k \leq K$, the splits are computed by minimizing the sum of squared errors of the following regression model:

$$\pi_{t+h} = \sum_{k=1}^K c_k I\{x_t \in R_k\}$$

$$I(x_t \in R_k) = \begin{cases} 1, & x_t \in R_k, \\ 0, & \text{otherwise.} \end{cases}$$

Where c_k is a constant that corresponds to the k th region, which is estimated as the sample average of the realizations of inflation falling into region R_k , i.e, $\hat{c}_k = \text{mean}(\pi_{t+h}^k)$ and $I(x_t \in R_k)$ is the indicator function. The final forecast is the average of the forecasts of each tree K_b applied to the original data:

$$\hat{\pi}_{t+h} = \frac{1}{B} \sum_{b=1}^B \left[\sum_{k=1}^{K_b} \hat{c}_k I\{x_t \in R_k\} \right]$$

Where $\hat{\pi}_{t+h}$ is the predicted h-step-ahead inflation. According to [Doojav and Narmandakh \(2025\)](#), the RF algorithm is able to balance the predictive accuracy with interpretability. It also excels at capturing nonlinear interactions, and these are crucial for understanding the complex dynamics of inflation. Random forest is also considered a reliable feature selection technique since Random Forest evaluates and extracts a random subset of features at each split, as advocated by [Aras and Lisboa \(2022\)](#). However, RF also has some limitations due to intensive and time-consuming computation ([Dewi and Chen \(2019\)](#)).

3.3.2 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is a decision tree-based ensemble algorithm using a gradient boosting setup, as proposed by [Chen and Guestrin \(2016\)](#). Its core idea is to iteratively add a weak learner (a decision tree) at a time to minimize the loss function. In other words, XGBoost builds decision trees sequentially, so each new tree is added to reduce the errors made by the previous one, and this process is also known as boosting. The XGBoost model can be mathematically represented as follows:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i)$$

where the XGBoost algorithm uses K additive functions to predict the output, $\hat{y}_i$ is the final predicted value, K is the number of trees, and $f_k(x_i)$ stands for the prediction of the k^{th} tree. XGBoost minimizes the following regularized objective function:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k)$$

Where $l(\hat{y}_i, y_i)$ is the differentiable loss function which measures the difference between the true value y_i and the predicted value $\hat{y}_i$, the second term $\Omega(f_k)$ is the regularization term that penalizes overly complex trees.

XGBoost is considered in this analysis because of the following reasons. One of them is to prevent the overfitting issue and improve the model's generalization by adding penalty terms to control the complexity of the trees. The second is that it uses parallel processing for faster training, given the large monthly U.S. macroeconomic and financial dataset. Therefore, XGBoost offers superior predictive performance in both generalization and speed. However, the shortcoming of this algorithm is that it has a large number of hyperparameters impacting the XGBoost's inflation predictive performance, as stated by [Aras and Lisboa \(2022\)](#).

3.4 Hybrid ML Models

The use of a hybrid inflation forecasting method is used to investigate whether the predictive accuracy of a given ML model is primarily determined by variable selection or the potential nonlinearities in the US inflation data. Inspired by the approach of [Medeiros et al. \(2021\)](#), we conduct the following steps:

1. Within each window, adaptive LASSO is employed to choose the most relevant predictors with non-zero coefficients. However, a safety condition is set to select the top two features with their largest absolute coefficient values in the case that adaLASSO selects fewer than two variables.
2. Using the set of top variables according to adaLASSO as input in the tree-based models: RF and XGBoost.
3. Tuning and estimating these ML models to forecast the US CPI inflation rate.

In this way, we have two hybrid ML models, including adaLASSO-RF and adaLASSO-XGBoost. Although the set of hybrid forecasting approaches can be further expanded, I believe that these two hybrid models are sufficient to test the relative importance of feature selection before model estimation and the nonlinearity in inflation dynamics for the ML forecasting performance. The results indicate that the variable selection technique does not improve the forecasting performance of ML models.

3.5 Statistical Test

3.5.1 Diebold-Mariano (DM) test

4 Data Description and Hyperparameter Tuning

5 Evaluation Results

5.1 Metrics

5.2 Variable Importance

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